

Intended Use

For the quantitative determination of direct bilirubin in serum on Hitachi analyzers. For *in vitro* diagnostic use only.

Method History

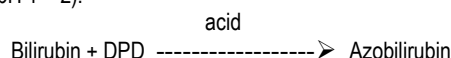
Since the introduction of the diazo method for bilirubin determination by Ehrlich in 1883,¹ several modifications have been proposed to enhance the reaction. The Malloy and Evelyn method² employs methanol to catalyze the azo-coupling reaction of the indirect Bilirubin, as well as to keep the azobilirubin in solution. A serious disadvantage of this method lies in the fact that protein may be precipitated by the methanol solution to yield falsely lowered results.

In 1938, Jendrassik and Grof.³ presented an assay that gave reliable results. The method is, however, cumbersome and involves several pipetting steps.

The method presented here was developed by Wahlefeld et al.⁴ The diazo reagent is 2,5-dichlorophenyldiazonium tetrafluoroborate (DPD) which reacts very rapidly in coupling with Bilirubin under acidic conditions. The resulting procedure is simple, yet exhibits good correlation when compared with the method of Jendrassik and Grof.

Principle

Direct Bilirubin is coupled with a diazonium salt (DPD) in a strongly acid medium (pH 1 – 2).



The intensity of the color of the azobilirubin produced is proportional to the Direct Bilirubin concentration and can be measured photometrically.

Reagents

1. Direct Bilirubin R1 reagent: acid buffer 50 mmol/L
2. Direct Bilirubin R2 reagent: acid buffer >30 mmol/L, >2.0 mmol/L DPD and stabilizers

Precautions

1. Reagents are toxic and corrosive. Do not pipette by mouth. Avoid contact with skin and clothing.
2. This reagent is for *in vitro* diagnostic use only.

Reagent Preparation

Reagents are supplied ready to use.

Reagent Storage

1. Packaged reagents may be stored at 2-8°C
2. Do not freeze reagents.
3. Avoid exposure to direct sunlight.

Reagent Deterioration

1. Do not use if reagents show evidence of microbial contamination (turbidity).
2. If the R2 develops very slight precipitation that re-dissolves when the R2 is warmed gently, the reagent may be used.
3. R2 reagent containing a precipitate that does not re-dissolve and results in product discoloration should not be used.
4. Do not use if reagent fails to achieve assigned assay values of fresh control sera.

Specimen Collection and Storage

1. Fresh, unhemolyzed serum is recommended.⁵
2. Samples should be analyzed within two hours of collection if kept at room temperature in the dark and within twelve hours if kept refrigerated (2-8°C) and protected from light.⁶
3. Bilirubin in serum is stable for three months when stored frozen (-20°C) and protected from light.⁶
4. Direct sunlight may cause up to a 50% decrease in bilirubin within one hour.⁷
5. Specimen collection should be carried out in accordance with NCCLS M29-T2. No method can offer complete assurance that human blood samples will not transmit infection. Therefore, all blood samples should be considered potentially infectious.

Interferences

1. All interference studies were performed according to the procedures recommended in NCCLS guideline No. EP7-P for interference testing in clinical chemistry.⁸
2. Serum hemoglobin levels up to 100 mg/dl do not interfere with results.
3. Serum Triglycerides up to 500 mg/dl do not interfere with results.
4. A number of drugs and substances affect bilirubin results. See Young, et al.⁹

Materials Provided

Direct bilirubin reagents R1 and R2.

Materials Required but not Provided

1. Controls
2. Calibrator
3. Hitachi analyzer
4. Application and instrument manuals

Procedure (Automated)(Hitachi 717)

TEST NAME	[DBIL]
ASSAY CODE	[2-POINT]:[24]-[30]
SAMPLE VOLUME	[6] [3]
R1 VOLUME	[250] [20] [NO]
R2 VOLUME	[65] [20] [NO]
WAVELENGTH	[660] [546]
CALIBRATION	[LINEAR] [0] [0]
STD (1) CONC-POS	[0.00] [1]
STD (2) CONC-POS	[*] [*]
STD (3) CONC-POS	
STD (4) CONC-POS	
STD (5) CONC-POS	
STD (6) CONC-POS	
SD LIMIT	[0.1]
DUPLICATE LIMIT	[200]
SENSITIVITY LIMIT	[0]
ABS LIMIT (INC/DEC)	[0] [INCREASE]
PROZONE LIMIT	[32000] [UPPER]
EXPECTED VALUE	[*] [*]
PANIC VALUE	[*] [*]
INSTRUMENT FACTOR	[1.0]

For other Hitachi instrument applications, please contact the manufacturer's Technical Service Department.

Calibration

Follow instrument application instructions for calibration. Refer to instrument manual instructions for calibration procedures and frequency. It is recommended that each laboratory determine its own frequency of calibration.

Quality Control

The validity of the reaction should be monitored by use of the control sera with known normal and abnormal direct bilirubin values. These controls should be run at least with every working shift in which direct bilirubin assays are performed. It is recommended that each laboratory establish its own frequency of control determination.

Calculations

Abs. = Absorbance
Unk. = Unknown
Cal. = Calibrator

$$\frac{\text{Abs. Unk.} - \text{Abs. Unk. Blank}}{\text{Abs. Cal.} - \text{Abs. Cal. Blank}} \times \text{Conc. of Cal. (mg/dl)} = \text{Direct Bilirubin (mg/dl)}$$

Sample: If Abs. of Unknown = 0.35, Abs. of Unknown Blank = 0.01, Abs. of Calibrator = 0.25, Abs. of Calibrator Blank = 0.01, Concentration of Calibrator = 4.0 mg/dl

Then:
$$\frac{0.35 - 0.01}{0.25 - 0.01} \times 4 = \frac{0.34}{0.24} \times 4 = 5.7 \text{ mg/dl}$$

Limitations

1. Samples with values above 10 mg/dl must be diluted 1:1 with isotonic saline, re-assayed and the final answer multiplied by two.
2. Serum hemoglobin levels of up to 100 mg/dl and triglycerides up to 500 mg/dl do not interfere with results.

Performance

The following performance data was obtained on a Hitachi 717 analyzer using the designated parameters and following standard operating protocols.

Sensitivity

Under the assay conditions described the sensitivity ($\Delta A/\text{mg/dl}$) of this method has been determined to be 0.086mg/dl.

Precision

Samples tested were commercial human Direct Bilirubin control samples. Studies were performed following a modification of NCCLS document EP5-T2.¹⁰ Within day results were derived from assaying 20 replicates of each sample in a single run. Day to Day results were derived from assaying each sample in duplicate, two times each day, for twenty days.

Within Day (n = 20)			Day to Day (n = 80)		
Mean	S.D.	C.V.%	Mean	S.D.	C.V.
2.1	0.042	2.01	1.9	0.088	4.57
5.5	0.074	1.35	5.4	0.082	1.51

Assay Range

0.0 – 10.0 mg/dl

Assay range was determined using commercially available linearity material.

Correlation

$Y = 1.02x + 0.13$

$r = 0.995$ (n=86, range = 0.0 – 21.4 mg/dl)

x = Roche Diagnostics Direct Bilirubin reagent.

Y = Pointe Scientific, Inc. Hitachi Direct Bilirubin reagent.

Correlation studies included human serum samples (including normal, elevated and "spiked" samples), commercial controls and commercial linearity material.

Expected Values (Direct)^{7,11}

Adults and infants (over one month): 0 – 0.5 mg/dl

It is strongly recommended that each laboratory establish its own normal range.

References

1. Ehrlich, P., Charite Ann. 8:140(1883).
2. Malloy, H.T., Evelyn, K.A., J. Biol. Chem. 119:481 (1937).
3. Jendrassik, L., Grof, P., Biochem. Zeitschr. 297:81 (1938).
4. Wahlefeld AW, et al. Scand J Clin Lab Invest. 29 Supplement 126(1972).
5. Michaelsson, M. Scand. J. Clin. Lab. Invest (Suppl. 49) 13:1 (1961)
6. Martinek, R.G., Clin. Chem. Acta 13:161 (1966).
7. Tietz, N.W. Fundamentals of Clinical Chemistry, Philadelphia, W.B. Saunders, P. 1028 (1976).
8. NCCLS document, "National Evaluation Protocols for Interference Testing", Evaluation Protocol Number 7, Vol. 4, No. 8, (June 1984).
9. Young, D.S., Effects of Preanalytical Variables on Clinical Laboratory Tests, Washington DC, AACC Press, (1997)
10. NCCLS document, "Evaluations of Precision Performance of Clinical Chemistry Devices", 2nd Ed. (1992)
11. Gambino, S.R., et al, Bilirubin Assay (Revised), Commission on Continuing Education, Am. Soc. of Clin. Path., Chicago, (1968).

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Symbols Key

 Use by (YYYY-MM-DD)	 LOT Lot and batch code	 REF Catalog number
 Manufacturer	 Temperature limitation	 Consult instructions for use
 IVD In vitro diagnostic medical device	 CE CE mark	
 EC REP	Authorized representative in the European Community	